

Run 06 | Fear of failure

India

IV + controls + NES

Target Fear of failure prevents starting a business; binary 1 vs. 0
Sample 27,276 respondents **Coverage** 1 country; 2015–2022
Positive cases 12,009 (44.03%) **Case time** 2m 05s
Train / val. / test 18,549 / 3,273 / 5,454 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business exit, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
NES controls (12): Financing, Government support, Taxes and bureaucracy, Government programs, School entrepreneurship education, Higher/vocational entrepreneurship education, R&D transfer, Commercial/legal/professional services, Market changes, Market entry, Physical infrastructure, Cultural and social norms. Total predictors: 47.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.7344). Highest F1: Random Forest (0.6637). For Random Forest, adding NES changed ROC-AUC by -0.0030 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.7344	0.6895	62.06	54.42	85.05	0.6637	3s
CatBoost	0.7239	0.6744	63.27	55.79	79.88	0.6570	1m 05s
XGBoost	0.7215	0.6684	60.07	52.83	86.71	0.6566	2s
H2O GBM	0.7210	0.6676	60.96	53.54	85.63	0.6589	6s
Ridge	0.6820	0.6247	54.86	49.30	90.09	0.6373	16s
Logistic Regression	0.6820	0.6245	55.65	49.79	89.13	0.6389	1s
Elastic Net	0.6813	0.6244	52.04	47.73	94.25	0.6337	8s
Lasso	0.6812	0.6243	51.94	47.68	94.25	0.6333	8s
Decision Tree	0.6588	0.5825	58.10	51.58	78.84	0.6236	<1s
Naive Bayes	0.6387	0.5529	52.07	47.68	91.09	0.6259	11s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 6

Rank	Predictor	Raw importance	Share%
1	Age	260.68	8.98
2	Household size	185.83	6.40
3	Occupational status	171.37	5.90
4	Good career choice	167.12	5.76
5	Opportunity perception	130.82	4.51
6	Household income	124.01	4.27

Share is the percentage of total forest impurity importance across all 88 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	28.20	0.97
2	Health	27.00	0.93

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Leading NES predictors within the same forest

Rank	NES predictor	Raw importance	Share%
1	Government programs	17.91	0.62
2	Taxes and bureaucracy	15.97	0.55
3	School entrepreneurship education	12.65	0.44

NES shares use total forest importance.

Source: Completed Objective 4 NES extension batch, 29 September 2026; 02_obj4_india_iv_cv_nes_fearfail_2015_2022. Scores and importance are taken directly from saved CSV outputs.